

the code snippet for the fingerprint enrolment process.

After successful enrolment, the system confirms the assigned name, maintaining traceability and identity mapping. Fingerprints may be enrolled for up to 15 individuals, as follows:

- 1: Administrator (Rakesh Jain)
- 2-5: Candidates (Anil, Gunmala, Rekha, Monu)
- 6-15: Voters (Niket through Deepak)

2. Fingerprint enrollment deletion.

If removal of a user is required, a separate deletion sketch allows a stored fingerprint to be erased from the module's memory. The serial monitor displays the name associated with the selected ID. When deletion is complete, a confirmation message appears. This function is particularly useful when preparing the system for a new election cycle.

Use this sketch to delete any enrolment by selecting an ID between 1 and 15. Fig. 4 shows the code snippet for this process.

(Note. The deletion sketch should be uploaded only when removal of a stored fingerprint is intended.)

3. Main voting code. The main voting code enables the actual voting process. The system waits for fingerprint input and checks it against the enrolment database. If a valid match is detected, the system verifies whether the individual has already voted. If the voter has already cast a vote, the attempt is denied. If not, the voter selects a candidate by pressing the

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corresponding button. Fig. 5 shows the snippet for main voting code.

When a vote is cast, the system records it and activates a buzzer connected to Pin 12 for one second, confirming successful vote registration. The fingerprint ID is then marked as 'voted,' preventing repeat voting. The system then returns to standby mode for the next user.

Final results are displayed on the LCD only when the administrator's fingerprint is detected. Pressing the RESET button on the Arduino Uno restarts the system and clears all recorded votes, enabling a new election cycle.

EFY note. Importantly, the system is not limited to 15 voters. With the current R305 fingerprint sensor, the system can be expanded, because the sensor internally stores 1000 fingerprint templates. The only required upgrade for larger-scale use is replacing Arduino Uno with Arduino Mega/NODE-MCU, which provides more memory for voter names and voting logic. This means the same system becomes suitable for schools, colleges, clubs, offices,

housing societies, and small institutional elections without redesigning the hardware.

Construction and testing

Begin by uploading the enrolment code and connecting the system to a computer. Open the serial monitor and follow the steps already described to complete fingerprint enrolment. If any stored fingerprint must be removed, upload the deletion code; otherwise, proceed to the next stage.

After enrolment, upload the main voting code to the Arduino Uno and assemble the hardware as shown in Fig. 6. After proper assembly, label each candidate's name beside the corresponding switch: Button 1 (pin 8) for Anil, Button 2 (pin 9) for Gunmala, Button 3 (pin A0) for Rekha, and Button 4 (pin A1) for Monu on the Arduino board. Once assembled, the system is ready for operation.

To cast a vote, the user places a finger on the sensor. When the fingerprint is validated, the buzzer sounds, and the vote can be cast using the corresponding push button. **EFY**

Rakesh Jain, Assistant Professor in ECE Department in Geetanjali Institute of Technical Studies, Udaipur, holds a master's degree in VLSI, BE degree in electronics and communication, and diploma in electronics. His research areas include sensors and microcontrollers. He has 31 copyrights, 10 registered design patents, 4 Indian utility patents, and 19 published DIY articles in EFY. He is recipient of Mewar Scientist Award 2023.



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